

A First Categorical DWLS Profile-Hessian Check

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Purpose

This note records a first diagnostic exploration of the regular profile-Hessian roadmap in a genuinely categorical DWLS setting. The goal is not a finite-sample simulation yet. The goal is to check whether the categorical DWLS weight channel can visibly change the nested-test reference law once we leave exact fit.

The companion script is

`docs/research/notes/ordinal_dwls_profile_check.R.`

1 Setup

The population is a four-variable binary ordinal latent-response model. The first-stage object is

$$x = (u, \gamma),$$

where u stacks the four thresholds and six tetrachoric correlations, and

$$\gamma = \text{diag}(\Gamma_u)$$

is the diagonal of the asymptotic covariance of u . DWLS uses $W = \text{diag}(\gamma)^{-1}$. The script computes u , γ , and the joint covariance of (u, γ) by finite-difference delta method from the 2^4 binary cell probabilities.

The nested pair is:

A : congeneric one-factor latent-response correlations, B : tau-equivalent one-factor latent-response

Thresholds are free in both models. Thus the nested contrast tests equality of the loading structure, not threshold equality.

For the nested contrast B versus A , the script compares three quadratic reference laws:

- U_{GN} : Gauss–Newton fixed-weight projector, using only u ,
- Q_{fixed} : fixed-weight profile Hessian in u ,
- Q_{full} : full profile Hessian in $x = (u, \gamma)$.

The last object is the categorical DWLS version of the general $Q_B - Q_A$ roadmap object.

2 Results

Scenario	$\sqrt{2f_A}$	$\sqrt{2f_B}$	$\text{tr}(Q_{\text{fixed}}\Gamma_u)$	$\text{tr}(Q_{\text{full}}\Gamma_x)$
Exact nested null	0.0000	0.0001	2.8949	2.8949
Near-null misspec., $\epsilon = 0.04$	0.0499	0.0499	2.9108	2.9850
Near-null misspec., $\epsilon = 0.08$	0.1003	0.1003	2.9709	3.1580
Restriction false, congeneric	0.0001	0.1261	2.9113	3.0918

The corresponding full/fixed trace ratios were

$$1.0000, \quad 1.0255, \quad 1.0630, \quad 1.0620.$$

The exact nested null behaves as expected: the residual is essentially zero, so the DWLS weight channel is dormant and the full profile law collapses onto the fixed-weight law. Under fixed misspecification, the full DWLS law separates from the fixed-weight law. In the two near-null misspecification directions, the trace increases by about 2.5% and 6.3%.

The eigenvalue movement is more informative than the trace alone. For $\epsilon = 0.08$, the leading eigenvalues changed from

$$(1.2945, 0.9387, 0.7377)$$

under the fixed-weight profile Hessian to

$$(1.5546, 0.9326, 0.6710)$$

under the full DWLS profile Hessian. So the weight channel does not merely rescale the law; it reshapes it.

3 Interpretation

This is consistent with the general theory. The DWLS profile score has two blocks:

$$s_u = -Wr, \quad (s_\gamma)_k = -\frac{1}{2} \frac{r_k^2}{\gamma_k^2}.$$

The γ block is residual-driven. It vanishes at exact fit and becomes a leading component under fixed misspecification. This is exactly why categorical DWLS is a strong applied example for the paper: the most common ordinal SEM workflow uses an estimated diagonal weight, and that weight is asymptotically irrelevant only in the exact-fit regime.

4 Caveats

The near-null misspecification scenarios were constructed to keep the congeneric and tau-equivalent fits nearly tied, but they are not yet an exact proof-level nested pseudo-null design. The reported score-gradient differences are small but not identically zero. The next pass should solve for misspecification directions that satisfy the nested pseudo-null conditions more exactly, then simulate finite-sample rejection rates under those population values.

The script is intentionally small and finite-difference based. It is useful for orientation, not for final numerical claims. A production version should reuse the analytic ordinal moment influence and DWLS weight machinery in `magmaan`.

5 Next Steps

1. Construct exact pseudo-null ordinal DWLS populations by solving the value and profile-gradient equality conditions for B and A .
2. Run finite-sample simulations comparing ordinary χ^2 , fixed-weight Q_{fixed} , and full DWLS Q_{full} references.
3. Repeat for multigroup threshold/loading invariance, where categorical DWLS is most common in practice.
4. Port the profile-Hessian calculation to `magmaan` using the existing ordinal moment and weight caches.